

Does Money Buy National Happiness? It Depends on Who You Ask

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Abstract

Is GDP per capita a good predictor of national well-being? Using all waves of the World Values Survey and the European Values Study (WVS/EVS, 1981–2023, 112 countries), I find that log GDP per capita explains on average 14% of the cross-country variance of eight well-being indicators, and that a country’s broad world region is a better predictor, in and out of sample, in 88% of 952 specifications. This advantage stems from Latin America and Eastern Europe, and it vanishes in the Gallup World Poll, where income predicts life evaluations as well as region. To understand why the two datasets disagree, I randomize the question wording (Cantril ladder vs. life satisfaction) and scale (0–10 vs. 1–10) in an online survey of 11,000 respondents in ten high-income countries. Asked to the same sample, the ladder is better predicted by GDP than life satisfaction (R^2 of 0.61 vs. 0.28 across countries), which reproduces about half of the Gallup–WVS difference in predictive capacity, but both questions yield income gradients as steep as Gallup’s, three times steeper than the WVS’s. Wording and samples thus both contribute to the discrepancy, and the new data lie midway between Gallup and the WVS.

Keywords: subjective well-being; life satisfaction; GDP; world regions; survey methodology; question wording.

JEL codes: I31; O15; O57; C83.

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1 Introduction

Which country is the happiest? The answer most often heard, based on the World Happiness Report, is a Scandinavian, high-income country. Behind this answer lies one of the most robust stylized facts of well-being research: across countries, average life evaluations increase with the logarithm of GDP per capita (Deaton, 2008; Sacks et al., 2012; Stevenson & Wolfers, 2008). Using the Gallup World Poll, Deaton (2008) finds a correlation above 0.8 between average life evaluation and log GDP per capita; using the World Values Survey (WVS), Inglehart & Klingemann (2000) reported an R^2 of about one half for an index of happiness and satisfaction. These correlations are frequently invoked to support the view that economic growth is a reliable route to higher well-being.

This paper revisits the predictive capacity of GDP per capita for national well-being, and asks two questions. First, how much of the cross-country variation in well-being does GDP per capita actually predict, and does it predict it better than a very simple alternative, the country's world region? Second, why do the two main sources of cross-national well-being data, Gallup and the WVS, give different answers to the first question? Previous work has noted that income is more strongly related to Gallup's ladder than to WVS measures, and attributed part of the difference to WVS samples in poor countries, which were mostly literate and urban (Deaton, 2008); it has also shown that national income is more related to life evaluations than to happiness or emotions (Inglehart et al., 2008; Kahneman & Deaton, 2010). Yet whether the difference between datasets comes from the question asked or from the samples has, to my knowledge, not been tested experimentally.

For the first question, I use all seven waves of the WVS, complemented with the 2017 wave of the European Values Study (342 country-years), eight indicators of national well-being, and seven income measures, and I compare the variance explained by income and by region using the Shapley (LMG) decomposition of R^2 (Grömping, 2007; Lindeman et al., 1980) and leave-one-country-out cross-validation. For the second question, I embedded a 2×2 experiment in an online survey of 11,000 respondents in ten high-income countries, randomizing the wording (Gallup's Cantril ladder vs. the WVS life-satisfaction question) and the scale (0–10 vs. 1–10), and I compare the relation between GDP and each question with the relation observed in Gallup and the WVS in the same countries.

The results are threefold. First, in the WVS, GDP per capita is a weak predictor of national well-being: it explains on average 14% of the variance of the eight indicators (20% with the best income measure), and nothing of the share of very happy people. Region is a better predictor in 88% of specifications and, out of sample, for every indicator.

Second, this conclusion is fragile in two respects: it is driven by Latin America and Eastern Europe, and it does not hold in Gallup, where log GDP per capita explains 61% of the cross-country variance of the mean ladder, against 48% for region. Third, the experiment shows that the question explains part, but not all, of this discrepancy. In the ten countries, log GDP explains 0.75 of the variance of Gallup’s ladder and 0.16 of WVS satisfaction. Asked to the same online sample, the ladder is also better predicted by GDP than the satisfaction question (R^2 of 0.61 vs. 0.28): wording accounts for about half (0.56) of the Gallup–WVS difference in predictive capacity, and the new data lie midway between the two datasets. However, both questions yield an income gradient as steep as Gallup’s (slopes of 3.60 and 3.86, vs. 4.20 in Gallup and 1.36 in the WVS): the satisfaction question adds dispersion around the income gradient, but the flat WVS gradient is specific to its samples. Wording also has a sizeable effect on levels: the ladder shifts answers by -0.45 points, which explains why Gallup levels are lower than WVS levels in high-income countries.

The paper contributes to three literatures. It adds to the debate on the relationship between income and well-being across countries and over time (Deaton, 2008; Easterlin, 1974; Easterlin et al., 2010; Jebb et al., 2018; Stevenson & Wolfers, 2008) by showing that the strength of the cross-country relationship depends on the dataset, and by benchmarking income against regions. It relates to work on regional and cultural differences in well-being, in particular the high well-being of Latin America (Graham & Lora, 2009; Rojas, 2016) and the low well-being of transition countries (Gurieva & Zhuravskaya, 2009), and on the comparability of well-being across cultures (Diener & Suh, 2000; Exton et al., 2015; Senik, 2014). Finally, it contributes to the methodological literature on the measurement of subjective well-being (Bond & Lang, 2019; Kahneman & Krueger, 2006; Kaiser & Oswald, 2022; OECD, 2013; Stone & Mackie, 2013), and specifically on the comparability of the Cantril ladder and life-satisfaction questions (Bjørnskov, 2010; Nilsson et al., 2024) and on context and mode effects (Deaton & Stone, 2016; Dolan & Kavetsos, 2016), by providing experimental estimates of wording and scale effects in ten countries.

Section 2 presents the data. Section 3 compares income and region as predictors of national well-being. Section 4 investigates the discrepancy between Gallup and the WVS. Section 5 discusses limitations, and Section 6 concludes.

2 Data

2.1 World Values Survey

The World Values Survey (WVS) is the longest-running cross-national survey of values and subjective well-being. I use the WVS time-series file (Haerpfer et al., 2022), which pools the seven waves conducted between 1981 and 2022, and complete it with the Joint EVS/WVS 2017–2022 dataset (EVS/WVS, 2024), which adds the 36 national surveys of the 2017 wave of the European Values Study (EVS, conducted in 2017–2021) and 2 recent WVS surveys (India 2023, Uzbekistan 2022). The EVS asks the same well-being questions, and the combination of the two surveys is standard practice (the so-called Integrated Values Surveys). The EVS and the WVS are distinct surveys: when both surveyed the same country the same year (3 cases), I keep both surveys as separate observations. For brevity, I refer to the combined data as the WVS: 501,911 respondents in 342 country-years and 112 countries. National samples are designed to be representative of the adult population, typically with 1,000 to 3,000 respondents, and most interviews are conducted face-to-face (though some recent national surveys used web, mail or mixed modes, e.g. the WVS in the United States in 2017 and in Japan in 2019, or the EVS in Switzerland in 2017). All statistics use the survey weights provided by the WVS and the EVS.

The WVS contains two questions on subjective well-being:

- *Happiness*: “Taking all things together, would you say you are: *Very happy*; *Quite happy*; *Not very happy*; *Not at all happy*.”
- *Life satisfaction*: “All things considered, how satisfied are you with your life as a whole these days?”, answered on a scale from 1 (*Completely dissatisfied*) to 10 (*Completely satisfied*).

Non-responses are rare (1.4% for satisfaction and 1.7% for happiness on average) and uncorrelated with GDP per capita (correlation: -0.09); they are excluded.

Indicators of national well-being. Because there is no consensus on how to aggregate ordinal answers into a national indicator, and because the ranking of countries can depend on this choice (Bond & Lang, 2019), I compute eight indicators for each country-year:

- *Happy*: share answering *Quite* or *Very happy*;
- *Very Happy*: share answering *Very happy*;

- *Very Unhappy*: share answering *Not at all happy*;
- $V. Happy - V. Unhappy$: difference between the two previous shares;
- *Happiness (mean)*: mean happiness, with answers coded +3, +1, -1, -3;
- *Satisfaction (mean)*: mean life satisfaction;
- *Satisfied*: share answering 6 to 10 to the life-satisfaction question;
- $Happy + Satisfied$: average of *Happy* and *Satisfied*, the index used by Inglehart & Klingemann (2000).

The indicators are positively but imperfectly correlated: across country-years, the correlation between *Happy* and *Satisfied* is 0.72, and between *Very Happy* and *Satisfaction (mean)* it is only 0.60 (Appendix Table A1).

2.2 Gallup World Poll and World Happiness Report

The Gallup World Poll interviews around 1,000 adults per country and year in more than 140 countries, by telephone where coverage is high (typically in high-income countries) and face-to-face elsewhere (Gallup, 2022). Its life-evaluation question is the Cantril ladder (Cantril, 1965): “Please imagine a ladder, with steps numbered from 0 at the bottom to 10 at the top. The top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step of the ladder would you say you personally feel you stand at this time?”

I use the distribution of answers by country and wave, tabulated from the Gallup World Poll microdata (2,351 country-years, 168 countries, 2006–2023), from which I compute the mean and the shares answering 6–10, 8–10, 9–10, 10, 0–4 and 0–2. These distributions are unweighted. To cover the most recent years, I also use the ladder means published in the World Happiness Report 2026 (World Happiness Report, 2026), which average Gallup data over three years up to 2023–2025.

2.3 Income

My benchmark income measure is the log of GDP per capita at purchasing power parity (PPP, constant 2021 international dollars) from the World Development Indicators (World Bank, 2026). As an alternative, I use GDP per capita at market exchange rates (constant 2015 US dollars). Missing values are filled in automatically, in three steps: (i) PPP series,

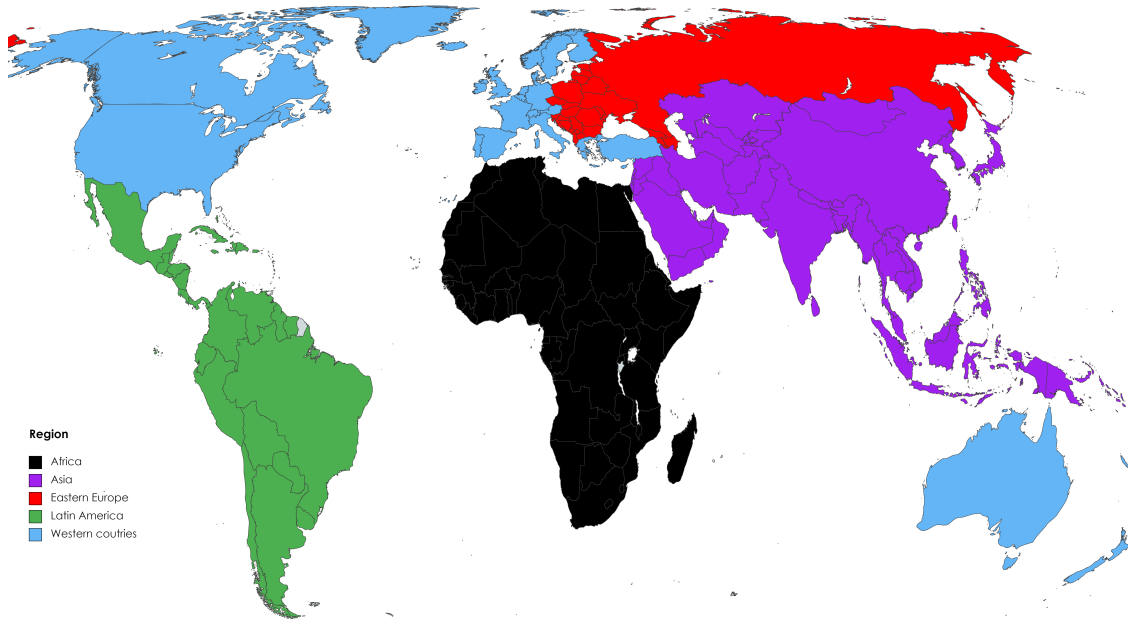
which start in 1990, are extended backwards (and gaps filled) using the growth rate of real GDP per capita; (ii) for countries absent from the World Bank data (e.g. Taiwan, or Venezuela in recent years), I use the IMF World Economic Outlook ([International Monetary Fund, 2026](#)), converted into the constant dollars of the World Bank series with a U.S. price deflator (the ratio of World Bank to IMF values for the United States); (iii) remaining gaps are filled with the closest year available (Montenegro 1996). Territories without national accounts use the GDP of the closest entity (e.g. the United Kingdom for Northern Ireland). A robustness check excludes all imputed values.

Region is a categorical variable, which can capture non-linear patterns that a single continuous variable cannot. To compare income and region on an equal footing, I also use categorical income indicators: income sextiles and income clusters obtained by k -means clustering of log GDP per capita, with $k = 5, 6$ or 7 , recomputed in each estimation sample. The clusters let the data choose the thresholds between income groups and allow for non-linear relationships.

2.4 Regions

My main classification groups countries into the five UN regional groups (Figure 1): *Africa*, *Asia* (the Asia-Pacific group, which includes the Middle East and Central Asia), *Eastern Europe* (the former Eastern Bloc, excluding Central Asia), *Latin America* (including the Caribbean), and *Western* countries (the “Western European and Others” group: Western Europe, North America, Australia, New Zealand, and Turkey). It departs from the UN groups for two countries only: Israel, a member of the Western group since 2000 because it was denied access to the Asia-Pacific group, is classified in Asia, and Cyprus, a member of the Asia-Pacific group, is classified with Western countries. Territories that are not UN members (e.g. Taiwan, Hong Kong, Puerto Rico, Kosovo) are assigned to the group of their neighbors. For robustness, I use four alternative classifications: the exact UN regional groups (with Israel in the Western group and Cyprus in Asia); a six-region variant separating the Middle East and grouping Central Asia with Eastern Europe; the seven World Bank regions (East Asia and Pacific, Europe and Central Asia, Latin America and the Caribbean, Middle East and North Africa, North America, South Asia, Sub-Saharan Africa), which isolate Latin America but group Eastern and Western Europe; and the five continents (Africa, Americas, Asia, Europe, Oceania), which isolate neither Latin America nor Eastern Europe.

Figure 1: Main classification of countries into five world regions.



2.5 A survey experiment on question wording and scale

Gallup and the WVS differ in many respects. Gallup asks the Cantril ladder on a 0–10 scale, the WVS life satisfaction on a 1–10 scale. Gallup relies on a centralized, standardized design, with about 1,000 respondents per country every year, interviewed by telephone in high-income countries and face-to-face elsewhere; the WVS is fielded every five years or so by national teams, mostly face-to-face, with sampling designs, modes and survey years that vary across countries. These differences fall into two categories that could explain why the two datasets disagree: the question (wording and scale) and everything else—sampling frames, interview modes, questionnaire context, translations, and survey years, which I refer to as *samples* in a broad sense. To separate the two, I embedded a randomized experiment in an original survey (Fabre, 2025). The survey was conducted online between April 15 and July 3, 2025, by Bilendi (or Kantar in Saudi Arabia) on 11,000 respondents in 10 high-income countries: the United States (3,000), Japan (2,000), Saudi Arabia (1,000), Germany (1,048), the United Kingdom (826), France (798), Italy (756), Spain (603), Poland (500) and Switzerland (469).¹ Samples were stratified with quotas on gender, age, income, education, region and urbanicity,² and results are reweighted to match population frequencies along these dimensions (with weights trimmed between 0.25 and 4).

¹The survey also covered Russia, where the well-being question was censored.

²Quota variables also included race in the U.S. and citizenship in Saudi Arabia.

Toward the end of the questionnaire, after questions on global policies, each respondent was randomly assigned to one of four versions of the life-evaluation question:

- *Ladder 0–10* (Gallup’s original): the Cantril ladder question quoted in Section 2.2, with steps numbered from 0 (*Worst possible*) to 10 (*Best possible*);
- *Ladder 1–10*: same wording, with steps numbered from 1 to 10;
- *Satisfaction 0–10*: the WVS question quoted in Section 2.1, with answers from 0 (*Completely dissatisfied*) to 10 (*Completely satisfied*);
- *Satisfaction 1–10* (WVS’s original): same wording, with answers from 1 to 10.

This 2×2 design identifies the effect of wording (ladder vs. satisfaction) and scale (0–10 vs. 1–10), and their interaction, holding the sample, mode, context and time constant. Within each country, each branch receives around 25% of respondents.

3 Income and region as predictors of national well-being

3.1 Empirical strategy

For each well-being indicator and each income indicator, I estimate three linear regressions across country-years i :

$$well-being_i = \alpha_1 + \beta_1 income_i + u_i \quad (1)$$

$$well-being_i = \alpha_2 + \gamma_2 region_i + e_i \quad (2)$$

$$well-being_i = \alpha_3 + \beta_3 income_i + \gamma_3 region_i + \varepsilon_i \quad (3)$$

where $region_i$ denotes the vector of region dummies. Denoting R_1^2 , R_2^2 and R_3^2 their coefficients of determination, the share of the variance explained by income alone is R_1^2 . To compare income and region, I decompose R_3^2 between the two regressors following Lindeman et al. (1980) (LMG), i.e. using the Shapley value (Grömping, 2007). With two (groups of) regressors, the LMG contribution of income is the average of its contribution when entered first (R_1^2) and when entered second ($R_3^2 - R_2^2$). The *share of explained variance due to income* is therefore

$$s = \frac{R_1^2 + (R_3^2 - R_2^2)}{2 R_3^2}, \quad (4)$$

and region is the better predictor whenever $s < 50\%$.

One feature of this comparison could favor region: region has four degrees of freedom while log GDP has one. I address this concern in three ways. First, I use categorical income variables (sextiles and clusters) with as many or more categories as regions. Second, I compute s with adjusted R^2 , which penalizes additional regressors; with more than 300 observations, the adjustment is small and only reinforces the conclusions. Third, and more directly relevant to *predictive* capacity, I compute leave-one-country-out cross-validated R^2 : each country's observations are predicted by a model estimated on all other countries. Unlike the in-sample R^2 , the cross-validated R^2 decreases when a model has superfluous parameters, and it is negative when a model predicts worse than the sample mean. Other robustness checks include each WVS wave separately, excluding the EVS surveys, weighting countries by their population, keeping only the last observation of each country, excluding the pandemic years 2020–2021, excluding imputed GDP values, using the previous vintage of GDP data (PPP in constant 2017 dollars), alternative region classifications, and excluding Latin America or Eastern Europe.

3.2 Income explains a small share of national well-being in the WVS

Figure 2 plots mean life satisfaction against GDP per capita for all WVS country-years. The relationship is positive, but the dispersion at any given income level is large: countries with a GDP per capita around 10,000 dollars span most of the range of observed life satisfaction, from the former Soviet republics at the bottom to Latin American countries at the top. Log GDP per capita explains 23% of the variance of mean satisfaction. The relation is even weaker for happiness (Figure 3). Appendix Figures A1–A7 plot the other indicators.

Table 1 generalizes this result to all indicators and income measures. On average over the eight indicators, log GDP per capita at PPP explains 14% of the variance, and the best-performing income measure (Cluster $k=7$ nominal) explains 20%. Satisfaction-based indicators are better explained than happiness-based ones: log GDP explains 4% of the variance of mean happiness and essentially none of the share of *Very happy* people, whose slope with respect to log GDP is statistically indistinguishable from zero (slope 0.002, $p = .948$; Appendix Table A2). This confirms the observation of Deaton (2008) and Inglehart et al. (2008) that income is more closely related to life evaluation than to happiness. By contrast, region alone explains 33% of the variance on average, and scores high even on happiness-based variables (last column).

Figure 2: Mean life satisfaction vs. GDP per capita (PPP), WVS/EVS 1981–2023.

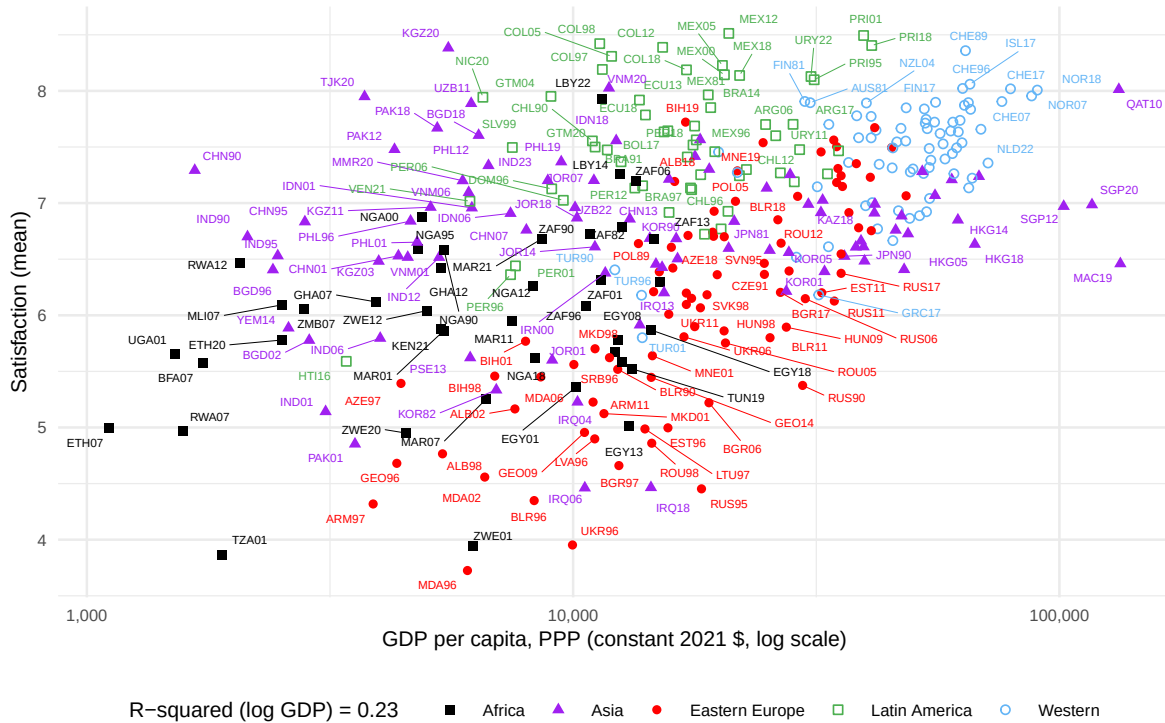
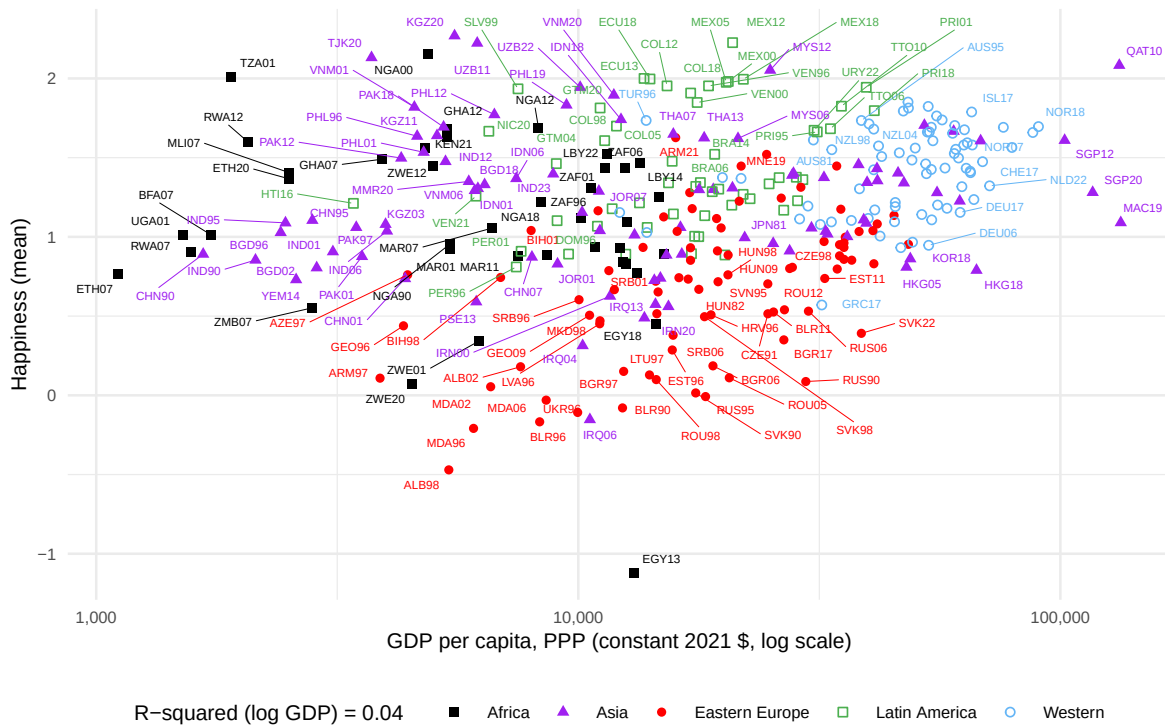


Figure 3: Mean happiness vs. GDP per capita (PPP), WVS/EVS 1981–2023.



Note: Answers coded +3 (Very happy), +1, -1, -3 (Not at all happy).

Table 1: Share of the variance of national well-being explained by income alone (R^2), WVS/EVS 1981–2023.

Well-being indicator	log GDP p.c.		Sextile PPP	Income cluster				Region
	PPP	nominal		$k=5$ PPP	$k=6$ PPP	$k=7$ PPP	$k=7$ nominal	
Happy	0.14	0.20	0.20	0.19	0.20	0.20	0.23	0.30
Very Happy	0.00	0.00	0.03	0.03	0.04	0.05	0.04	0.33
Very Unhappy	0.07	0.10	0.12	0.10	0.13	0.13	0.14	0.14
V. Happy – V. Unhappy	0.00	0.01	0.04	0.04	0.04	0.05	0.05	0.33
Happiness (mean)	0.04	0.08	0.10	0.08	0.09	0.10	0.11	0.33
Satisfaction (mean)	0.23	0.29	0.23	0.21	0.22	0.23	0.29	0.41
Satisfied	0.32	0.38	0.32	0.30	0.32	0.32	0.38	0.39
Happy + Satisfied	0.27	0.35	0.30	0.28	0.30	0.30	0.35	0.38
Mean	0.14	0.18	0.17	0.15	0.17	0.17	0.20	0.33
Max	0.32	0.38	0.32	0.30	0.32	0.32	0.38	0.41

Note: R^2 of the regression of each well-being indicator (rows) on each income indicator (columns), over 342 WVS/EVS country-years (a few with missing GDP are excluded). The last column gives the R^2 of the regression on region dummies. The replication files include an Excel version of this table with the R^2 of all region classifications (tables/main/r2_income_all_regions.xlsx).

3.3 Region predicts national well-being better than income in the WVS

Table 2 reports the share of explained variance due to income, s , for each indicator and income measure. It is below 50% in all cases but one, where income and region are tied (0.50). On average, the best income measure accounts for 33% of the variance explained jointly by income and region. The first split of a regression tree using log GDP and region as candidate variables is on region for all national well-being indicators.

The result is not an artifact of the larger number of parameters of the region model. Out of sample, region predicts better than income for every indicator (Appendix Table A3): for mean satisfaction, the leave-one-country-out R^2 is 0.37 with region against 0.21 with log GDP, and for several happiness indicators, income has no out-of-sample predictive power at all (negative cross-validated R^2). Across the 952 estimates of the robustness checks (indicator $\times$ income measure $\times$ specification; Appendix Table A4), region is the better predictor in 88% of cases (79% with the best income measure). The result holds in each wave except wave 4 (1999–2004, 40 countries), where income and region perform similarly; when countries are weighted by population; without the EVS surveys; without the pandemic years; without imputed GDP values; and with the previous vintage of GDP data.

Table 2: Share of explained variance due to income rather than region (LMG), WVS/EVS 1981–2023.

Well-being indicator	log GDP p.c.		Sextile PPP	Income cluster			
	PPP	nominal		k=5 PPP	k=6 PPP	k=7 PPP	k=7 nominal
Happy	0.29	0.36	0.38	0.35	0.37	0.38	0.41
Very Happy	0.00	0.01	0.07	0.06	0.08	0.10	0.10
Very Unhappy	0.28	0.38	0.43	0.40	0.47	0.47	0.50
V. Happy – V. Unhappy	0.01	0.02	0.09	0.07	0.09	0.11	0.11
Happiness (mean)	0.09	0.13	0.18	0.14	0.17	0.19	0.19
Satisfaction (mean)	0.32	0.38	0.32	0.29	0.31	0.32	0.38
Satisfied	0.43	0.49	0.43	0.41	0.43	0.44	0.49
Happy + Satisfied	0.39	0.46	0.42	0.39	0.41	0.42	0.47
Mean	0.23	0.28	0.29	0.26	0.29	0.30	0.33
Max	0.43	0.49	0.43	0.41	0.47	0.47	0.50

Note: Share of the R^2 of Equation (3) attributed to income by the LMG (Shapley) decomposition, Equation (4). Values below 0.5 indicate that region is the better predictor. Values above 0.5 (income is the better predictor) are in bold.

3.4 Where the region advantage comes from

Table 3 shows that the region advantage rests on two well-known “anomalies”: Latin America, happier than its income would predict (Graham & Lora, 2009; Rojas, 2016), and Eastern Europe, unhappier (Guriev & Zhuravskaya, 2009). Region remains the better predictor when either one is excluded (in 70% of cases without Latin America and 73% without Eastern Europe), but not when both are (27%). Consistently, the result holds with classifications that isolate both groups (exact UN groups and six regions: 100%), weakens with the World Bank regions, which isolate Latin America but merge Eastern and Western Europe (55%), and disappears with continents, which isolate neither (48%).

Most importantly, the conclusion does not carry over to Gallup data (Figure 4 and Table 4). In Gallup’s latest observation for each of 168 countries, log GDP explains 61% of the variance of the mean ladder, against 48% for region, and income accounts for 60% of the explained variance. Income is the better predictor for the mean and for the shares of respondents in the middle and bottom of the distribution, with all income measures. Over all Gallup specifications, region is the better predictor in only 34% of cases, mostly for indicators of the upper tail of the distribution (shares answering 9–10 or 10), which are almost unrelated to income—echoing the weak relation between income and the share of *Very happy* people in the WVS. With the latest WHR data (2023–2025), log GDP explains 63% of the variance of the ladder against 55% for region. Whether “GDP is a poor predictor of national well-being” therefore depends on which of the two datasets one trusts, a question I turn to in Section 4.

Table 4: Share of explained variance due to income rather than region (LMG), Gallup, last observation per country.

Well-being indicator	log GDP p.c.		Sextile PPP	Income cluster			
	PPP	nominal		k=5 PPP	k=6 PPP	k=7 PPP	k=7 nominal
Satisfied	0.58	0.61	0.59	0.58	0.60	0.59	0.60
Very satisfied (8–10)	0.44	0.51	0.49	0.47	0.49	0.51	0.53
Extremely satisfied (9–10)	0.16	0.25	0.24	0.27	0.22	0.30	0.28
Completely satisfied (10)	0.02	0.02	0.08	0.09	0.06	0.08	0.08
Unsatisfied (0/1–4)	0.61	0.60	0.60	0.59	0.61	0.60	0.59
Dissatisfied (0/1–2)	0.66	0.63	0.67	0.66	0.67	0.67	0.65
Satisfaction (mean)	0.60	0.62	0.59	0.58	0.60	0.59	0.60
Mean	0.44	0.46	0.47	0.46	0.47	0.48	0.48
Max	0.66	0.63	0.67	0.66	0.67	0.67	0.65

Note: 168 countries, last Gallup observation available (2006–2023). Same definitions as Table 2. Values above 0.5 (income is the better predictor) are in bold.

3.5 Other correlates, within-country evidence and happiest countries

What do regions capture? The average perceived freedom of choice and control over one’s life alone explains 63% of the variance of mean satisfaction; in a Shapley decomposition with income and region, it accounts for 42 percentage points, against 21 for region and 11 for income (Appendix Table A5). This variable is itself a subjective evaluation, so the correlation should not be read causally, but it suggests that non-material dimensions of life are central to regional differences (Inglehart et al., 2008). Tolerance of homosexuality (24%), the importance of God (1%) and GDP growth over the previous five years (2%) explain much less.

The cross-sectional relation should not be confused with the relation over time. Within countries (country fixed effects, 312 observations), mean satisfaction increases with log GDP per capita, with a slope of 2.31 per unit of $\log_{10}$ GDP, larger than the cross-sectional slope (1.14), although income growth explains only 0.34 of the within-country variance (Appendix Table A6). These associations are consistent with Stevenson & Wolfers (2008) and Sacks et al. (2012) rather than with a strict version of the Easterlin paradox (Easterlin, 1974; Easterlin et al., 2010), although they may reflect common shocks (e.g. transition recessions) rather than the effect of income per se.

Finally, rankings of countries depend on the indicator. Counting, for each indicator and each wave (and all waves combined), the happiest country-year, Switzerland appears most often (10 times out of 64), followed by Mexico (9) and Kyrgyzstan (6) (Appendix Table A7). The happiest country-years are Western (25), Latin American (19), Asian (15) or African (5). Similarly, with Gallup data on emotions, Blanchflower & Bryson (2024)

find that the top-ranked country or U.S. state differs across measures: Denmark for the ladder, Hawaii for enjoyment, Paraguay for smiling, Bhutan for being well-rested, and Vietnam, Taiwan, Uzbekistan and Finland for the lowest prevalence of pain, sadness, worry and anger, respectively. The link between income and well-being seems to be that rich countries rarely experience low well-being, rather than that they reach the highest levels.

4 Why do Gallup and the WVS disagree? Wording vs. samples

4.1 The discrepancy

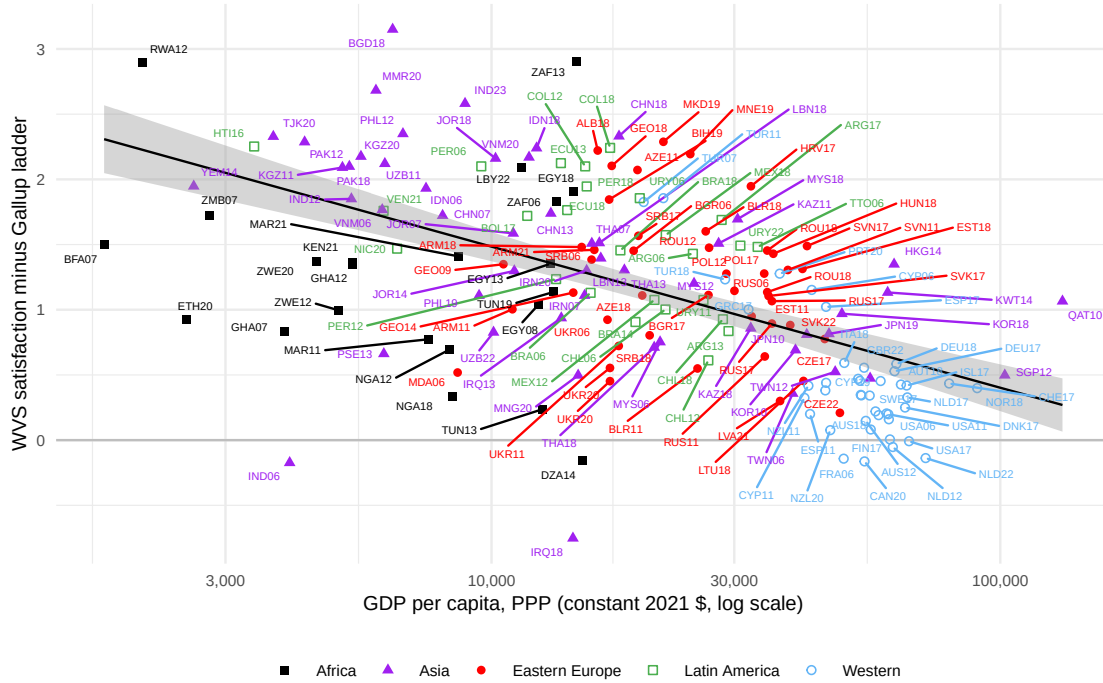
Restricting both datasets to the 187 country-years (99 countries) observed by both surveys the same year, log GDP explains 0.53 of the variance of Gallup’s mean ladder but only 0.16 of WVS mean satisfaction, and income accounts for 51% of the variance explained jointly with region in Gallup against 24% in the WVS. The discrepancy found in Section 3 is thus not due to different country coverage. The gap between the two datasets decreases strongly with GDP (Figure 5): the slope is -1.08 per unit of $\log_{10}$ GDP (s.e. 0.14, clustered by country), with WVS answers exceeding Gallup answers by around two points in low-income countries, against less than half a point in high-income countries.

The two datasets differ in their question (ladder 0–10 vs. satisfaction 1–10) and in their samples in the broad sense defined in Section 2.5. The experiment allows me to measure how much of the discrepancy the question accounts for: first for the relation between income and well-being, which is the object of this paper (Section 4.2), then for levels (Section 4.3).

4.2 Does wording explain the discrepancy? Evidence from ten countries

The ten countries of the experiment provide a direct test. Since the object of interest is national well-being, the unit of observation is the country: for each question variant, I compute the mean answer in each country and regress it on log GDP per capita across the ten countries, as for the Gallup and WVS country means (Table 5 and Figure 6). Each country thus has the same weight, whatever its sample size. For Gallup and the WVS, I use the latest WVS or EVS survey of each country (2017–2022, except Saudi Arabia, 2003)

Figure 5: WVS satisfaction minus Gallup ladder in the same country-year, by GDP per capita.



Note: Mean answers on native scales (WVS: 1–10; Gallup: 0–10), 187 country-years observed by both surveys. Line: OLS fit with 95% confidence band.

and the Gallup data of the same year (or the closest year, at most three years apart). Confidence intervals come from a bootstrap that resamples respondents in the three surveys.

I first compare R^2 , which directly measure the predictive capacity of GDP per capita. In the past data, the ten countries reproduce the global discrepancy: log GDP explains 0.75 (95% CI [0.68; 0.80]) of the cross-country variance of Gallup's ladder but only 0.16 ([0.10; 0.24]) of WVS satisfaction. Asked to the same online sample, the two original questions reproduce much of this contrast: log GDP explains 0.76 ([0.52; 0.83]) of the variance of the ladder 0–10 but 0.31 ([0.16; 0.45]) of satisfaction 1–10. Pooling both scales of each wording, the R^2 is 0.61 for the ladder and 0.28 for satisfaction. The wording thus accounts for 0.56 (95% CI [0.22; 0.82]) of the Gallup–WVS difference in R^2 (0.76 with the two original variants only, [0.31; 1.07]), and samples for the rest. In other words, the new data lie midway between Gallup and the WVS. Pooling the four variants, log GDP explains 0.43 of the variance, and income accounts for 0.58 of the variance explained jointly with region, halfway between Gallup (0.78) and the WVS (0.35). The two crossed variants, which combine the wording of one survey with the scale of the other (ladder 1–10 and satisfaction 0–10), also yield intermediate results, with R^2 of 0.33 and 0.23 and slopes of 2.57 and 3.21. Only Gallup's original variant yields a high R^2 , so that with ten

countries, the effect of the wording cannot be clearly separated from that of the scale.

Table 5: Relation between GDP per capita and national well-being in the ten countries of the experiment: past vs. new data.

Data	R^2 income	95% CI	Slope on $\log_{10}$ GDP	s.e.	R^2 region	Share due to income
Gallup (ladder 0–10)	0.75	[0.68; 0.80]	4.20	0.80	0.29	0.78
WVS/EVS (satisfaction 1–10)	0.16	[0.10; 0.24]	1.36	1.44	0.34	0.35
Fabre 2025: ladder 0–10	0.76	[0.52; 0.83]	4.35	1.05	0.14	0.91
Fabre 2025: ladder 1–10	0.33		2.57	1.36	0.50	0.36
Fabre 2025: satisfaction 0–10	0.23		3.21	2.03	0.36	0.35
Fabre 2025: satisfaction 1–10	0.31	[0.16; 0.45]	4.05	2.09	0.29	0.52
Fabre 2025: ladder, both scales	0.61	[0.44; 0.70]	3.60	1.23	0.33	0.71
Fabre 2025: satisfaction, both scales	0.28	[0.16; 0.37]	3.86	2.14	0.32	0.45
Fabre 2025: all four variants	0.43	[0.32; 0.51]	3.73	1.55	0.34	0.58

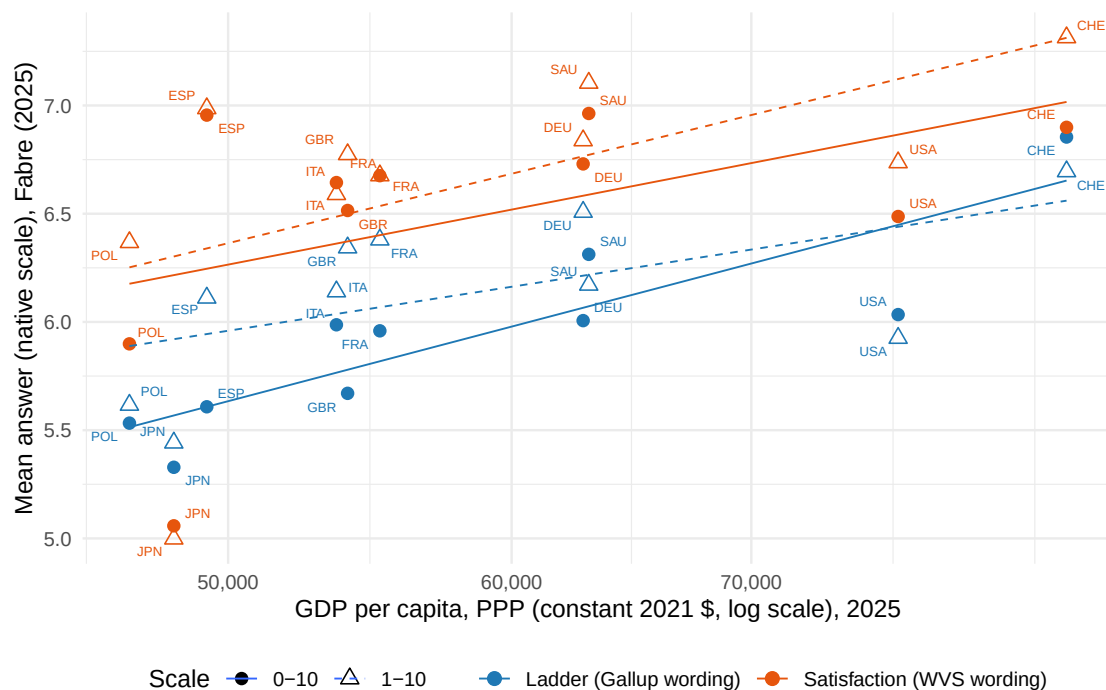
Note: OLS across the 10 countries of the experiment, one observation per country. Gallup and WVS/EVS: latest WVS/EVS survey and Gallup the same year, GDP of that year. Fabre (2025): mean answer by variant (native scale), GDP of 2025; “both scales” and “all four variants”: average of the variant means, 1–10 answers being linearly stretched to 0–10. 95% CI of R^2 : bootstrap resampling respondents. Standard errors of slopes robust to heteroskedasticity (HC1). R^2 region and share due to income: see Section 3.1 (regions: Western, Asia, Eastern Europe).

Slopes tell a different story. Both questions yield an income gradient close to Gallup’s (4.20): 4.35 with the ladder and 4.05 with the satisfaction question (3.60 and 3.86 when pooling scales), whereas the WVS gradient is three times flatter (1.36). The part of the Gallup–WVS difference in slopes that is not due to the question is 2.54 (95% CI [0.90; 4.18]). How can the satisfaction question yield a lower R^2 but the same slope as the ladder? With a single regressor, $R^2 = \beta^2 \text{var}(x) / \text{var}(y)$: for a given slope, the R^2 is lower when country means are more dispersed around the income gradient. The satisfaction question does not weaken the gradient, but it produces larger country-specific deviations from it—most visibly in Japan, whose mean satisfaction (5.00 on the 1–10 scale) is 1.4 points below that of any other country despite its high income, whereas its mean ladder (5.33) is only 0.2 points below the next lowest (Appendix Table A9). These country-specific deviations are precisely what region captures in the WVS. By contrast, the flat WVS gradient is specific to its samples.

I give more weight to the R^2 than to the slopes, because the question of this paper is the predictive capacity of GDP per capita, which the R^2 measures. Taken together, the results suggest that the wording accounts for about half of the difference in predictive capacity between Gallup and the WVS, through larger country-specific deviations in answers to the satisfaction question, while differences between samples account for the other half and for most of the difference in the income gradient. Two caveats apply. First, the bootstrap confidence intervals reflect the sampling of respondents and treat the ten countries as given; with only ten countries, R^2 and slopes are imprecise (Table 5 reports the stan-

dard errors of slopes across countries) and sensitive to individual countries. For instance, excluding Japan raises the R^2 of Gallup to 0.91 and lowers the share of the Gallup–WVS difference in R^2 due to wording to 0.38. Second, the decompositions between income and region are of limited use here, since seven out of the ten countries are Western.

Figure 6: Mean life evaluation vs. GDP per capita (PPP), by question variant, Fabre (2025).



Note: Mean answer on the native scale of each variant (0–10 or 1–10), ten countries surveyed in 2025. Colors: wording; shapes and line types: scale. Lines: OLS fits by variant.

Can the question explain the global pattern, where the gap between WVS and Gallup answers is largest in low-income countries? To gauge this, I estimate the question effect as a linear function of log GDP across the ten countries, and I convert each WVS country mean into a ladder-equivalent by subtracting the question effect predicted for its GDP. As the question effect is not significantly related to GDP within the experiment (slope -0.30 , s.e. 1.75), which is unsurprising given the narrow range of GDP among the ten countries, this adjustment raises the R^2 of WVS satisfaction on log GDP only from 0.16 to 0.26, far from Gallup’s 0.53. This adjustment only captures level differences that vary with GDP, not the additional country-specific dispersion that the satisfaction question produces in the experiment. Moreover, the experiment cannot rule out that the wording effect is much larger in low-income countries—for instance if the “best possible life” evokes material comparisons more strongly where material deprivation is widespread. Nothing in the high-income evidence points in that direction, though: within countries, the ladder does

not steepen the individual income gradient (Section 4.3).

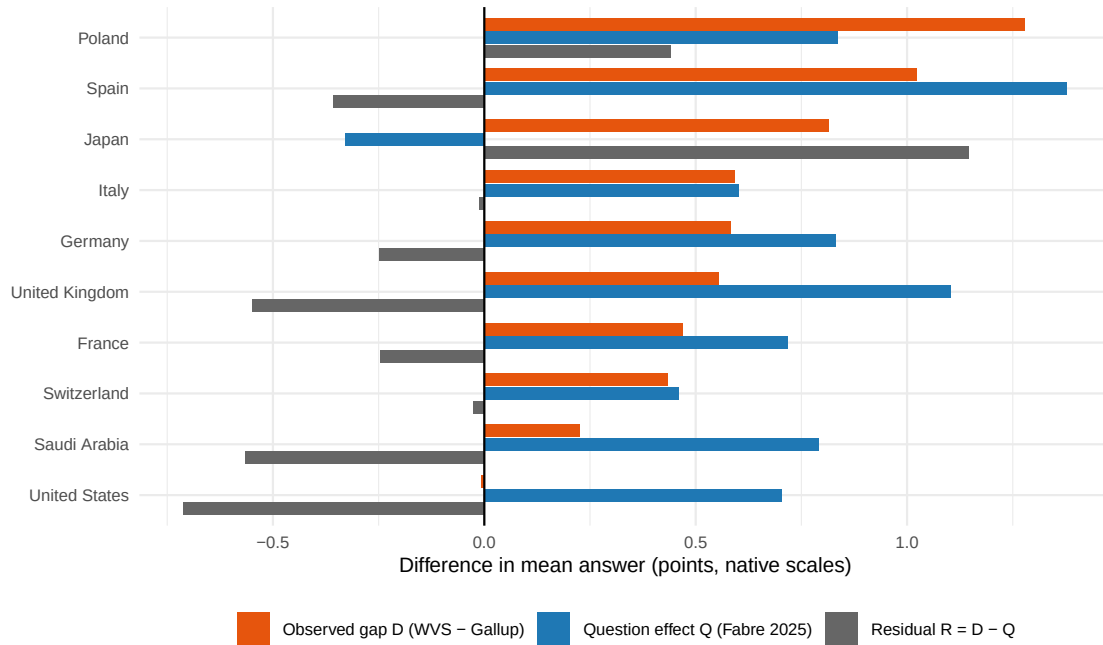
4.3 Wording and the levels of well-being

Although levels matter less than correlations for our question, the experiment also explains why the WVS yields higher levels than Gallup. Appendix Table A8 reports regressions of the answer on the randomized treatments, with country fixed effects, survey weights and heteroskedasticity-robust standard errors. The ladder wording shifts answers by -0.45 points on average (s.e. 0.05): respondents rate their lives lower on the “best possible life” ladder than when asked how satisfied they are, in line with evidence that the ladder evokes power and wealth more than life satisfaction does (Nilsson et al., 2024). Answers on the 1–10 scale are 0.14 points higher than on the 0–10 scale (s.e. 0.05). Once 1–10 answers are linearly stretched to 0–10 with $x \mapsto (x - 1) \times 10/9$, which maps 1 to 0 and 10 to 10, the scale effect becomes -0.26 : the stretch lowers each answer x by $(10 - x)/9$, i.e. by about 0.4 points at the average answer. Wording and scale do not interact. The wording effect varies markedly across countries (Figure 7; $\chi^2(9) = 93.4$, $p < .001$). Japan is the exception: it is the only country where the ladder yields *higher* answers than the satisfaction question (by 0.36 points); elsewhere, the wording effect ranges from -1.12 (Spain) to -0.34 points. Finally, the ladder does not steepen the income gradient *within* countries: the interaction between the ladder wording and the respondent’s income quartile is -0.004 ($p = .919$), against a gradient of 0.69 points per quartile.

To relate these effects to the Gallup–WVS gap, I define for each country c the observed gap $D_c = W_c - G_c$, where W_c is mean satisfaction in the WVS and G_c the mean ladder in Gallup the same year (as in Section 4.2), and the question effect $Q_c = F_c(\text{satisfaction 1–10}) - F_c(\text{ladder 0–10})$, the difference between the mean answers to the two original variants in the experiment, measured on the same sample. The residual $R_c = D_c - Q_c$ is the part of the gap due to samples in the broad sense. On average, WVS answers exceed Gallup answers by $D = 0.60$ points, and the question effect is even larger: $Q = 0.71$ points (Appendix Tables A9 and A10). The question thus fully accounts for the higher level of WVS answers in high-income countries, and the residual is small ($R = -0.11$, 95% CI $[-0.26; 0.04]$). Since the two components move in opposite directions, another way to summarize the level difference is to compare the size of the two movements: the question accounts for $|\bar{Q}| / (|\bar{Q}| + |\bar{R}|) = 0.86$ (95% CI $[0.76; 0.99]$) of the gross difference in levels.

The question effect does not, however, explain why the gap varies across countries. Since $D_c = Q_c + R_c$, the cross-country variance of the gap splits into $\text{var}(D) = \text{cov}(D, Q) + \text{cov}(D, R)$. The share attributable to the question, $\text{cov}(D, Q) / \text{var}(D)$, is only 0.11 (95% CI

Figure 7: Observed WVS–Gallup gap, question effect and residual by country.



Note: D : mean satisfaction (WVS/EVS, 1–10) minus mean ladder (Gallup, 0–10), same year. Q : mean answer to satisfaction 1–10 minus mean answer to ladder 0–10, Fabre (2025). $R = D - Q$.

$[-0.33; 0.55]$): countries with a larger question effect do not have a larger gap. Accordingly, predicting each country’s Gallup mean by subtracting its own question effect Q_c from its WVS mean does worse (root mean squared error: 0.54) than simply subtracting the average gap (0.35). Samples asked the same question also differ markedly. Gallup respondents (2023–2025) rate their lives 0.75 points higher on the ladder 0–10 than online respondents in 2025, and WVS respondents 0.75 points higher on satisfaction 1–10 (even in the United States, where the 2017 WVS was itself conducted online: 0.49 points). And in the 10 countries surveyed separately by the EVS and the WVS in 2017–2022, with the same question, mean satisfaction differs by 0.28 points on average (up to 0.53).

4.4 Which dataset has more representative samples?

If samples matter, which dataset is closer to the truth? There is no clear-cut argument in favor of either dataset, nor of the online sample used in the experiment. Gallup applies a centralized, standardized design: probability samples of about 1,000 adults per year, drawn by random-digit dialing where telephone coverage is high and by multistage area sampling elsewhere, with random selection of the respondent within the household and post-stratification weights on gender, age and, where possible, education (Gallup, 2023). But its samples are small; its face-to-face samples select households by random

route, with replacement of non-responding households; response rates are not published; interviews switched from face-to-face to telephone during the pandemic; and in 2023, a fifth of the interviews in 27 high-income countries came from web panels, most of them opt-in (Gallup, 2023). Its subgroup estimates have also been questioned (Blanchflower, 2024). Besides, Gallup microdata are not public, so that the representativeness of its samples cannot be assessed independently; lacking quotas, its samples could under-represent hard-to-reach groups such as less educated or rural people. The WVS requires probability samples of at least 1,200 respondents but allows quota selection at the last stage, and its sampling designs, modes and survey years vary across countries (World Values Survey Association, 2023). Tests flagging near-duplicate interviews (pairs of respondents giving identical answers to almost all questions, a possible sign of fabrication) in international survey programs (Kuriakose & Robbins, 2016) have been disputed (Pew Research Center, 2016). Finally, online panels exclude people without internet access, and quotas on observable characteristics do not guarantee representativeness in unobservable ones.

I test the representativeness of WVS/EVS samples directly (Appendix Table A11). Compared with population benchmarks (Barro & Lee, 2013; World Bank, 2026), WVS samples over-represent tertiary-educated adults (aged 25 and over) by a median factor of 1.98 in countries with a GDP per capita below 10,000 dollars, against 1.05 in countries above 30,000 dollars (295 surveys). They also over-represent urban residents in poorer countries (median ratio 2.19 vs. 1.22) and under-represent people aged 65 or more (0.64 vs. 1.01). The internal criterion of Kohler (2007)—women should make up half of married or cohabiting respondents—is violated at the 5% level in 36% of surveys. WVS samples are thus less representative in poorer countries, as noted by Deaton (2008), which is where their gap with Gallup is largest. However, correcting these observable imbalances has little effect. Controlling for log GDP, the over-representation of the tertiary-educated does not predict the gap (coefficient -0.07 , s.e. 0.07 , $p = .311$), and reweighting each WVS sample to the benchmark share of tertiary-educated adults barely changes its relation with income: in the country-years also covered by Gallup, the R^2 of mean satisfaction on log GDP goes from 0.19 to 0.20, against 0.54 for Gallup. The discrepancy thus stems from unobserved differences in who responds, how, and in which context, which the available evidence does not allow me to attribute to one dataset rather than the other.

5 Discussion

5.1 Income vs. region

Region is not a mechanism: it bundles culture, history, institutions, social ties, and possibly cross-cultural differences in how people use response scales (Chen et al., 1995; Diener et al., 2003; Exton et al., 2015). If regional differences partly reflect response styles, region predicts *measured* rather than *experienced* well-being. Studies of migrants suggest that national differences partly reflect a cultural determinant of well-being rather than translation artifacts (Exton et al., 2015; Senik, 2014), but the present analysis cannot separate these interpretations; anchoring vignettes (Kapteyn et al., 2010) or measures of emotional well-being would help. Moreover, the advantage of region rests on Latin America and Eastern Europe. The fairest summary is that GDP per capita leaves much of the cross-country variation unexplained, and that a few large regional deviations explain more of it than income does in the WVS.

The modest cross-country role of income echoes within-country evidence. Studies comparing people within countries find a small effect of income on life satisfaction relative to other circumstances such as employment or marital status (Blanchflower & Oswald, 2004; Helliwell, 2003); Kahneman et al. (2006) argue that even these estimates overstate the role of income, because of a “focusing illusion” when people evaluate their lives; and in German panel data, life satisfaction adapts to income increases within a few years (Di Tella et al., 2010). Conversely, emotional well-being increases with log income for most people in the United States (Killingsworth et al., 2023), and the within-country association between GDP and satisfaction documented above shows that the weak cross-sectional relation does not imply that income is irrelevant.

5.2 Wording vs. samples

The experiment shows that the question matters in two ways. It matters for levels: the Cantril ladder yields lower answers than the WVS life-satisfaction question, and this difference fully accounts for the lower average level of Gallup data in high-income countries. And it matters for predictive capacity: asked to the same sample, the ladder is better predicted by GDP, because the satisfaction question produces larger country-specific deviations from the income gradient. This reproduces about half of the Gallup–WVS difference in R^2 . But the question does not explain the flat income gradient of the WVS, nor the cross-country pattern of the Gallup–WVS gap in levels, which reflect differences between samples in the broad sense: sampling frames, interview modes (Conti & Pud-

ney, 2011; Dolan & Kavetsos, 2016), questionnaire context (Deaton, 2012; Deaton & Stone, 2016), translations, and time.

Several limitations should be kept in mind. First, the experiment covers only ten high-income countries, whereas the gap varies most across income levels, and where both Gallup and the WVS mostly interview face-to-face; only an experiment in low- and middle-income countries can settle this. Second, the residual lumps together several factors that the design cannot separate; in particular, the question effect was measured online, toward the end of the questionnaire, and may differ from the effect that would be obtained at the beginning of an interview conducted face-to-face or by phone. Third, the comparison with past data assumes that the question effect measured in 2025 applies to the surveys of 2017–2022 (2003 for Saudi Arabia).

6 Conclusion

GDP per capita is a weaker predictor of national well-being than is often assumed, but how weak depends on the data. In the WVS, it explains a small share of the cross-country variance of life satisfaction, less than a country's world region, and an even smaller share of self-declared happiness. The richest countries are not necessarily the happiest: the country-year with the highest share of very happy people in the WVS is Kyrgyzstan in 2020 (68%, with a mean life satisfaction of 8.4), with a GDP per capita of about 5,500 dollars; and many middle-income countries, especially in Latin America, are as satisfied as the richest ones. In the Gallup World Poll, however, income predicts life evaluations as well as region does. A survey experiment in ten high-income countries shows that both the question and the samples contribute to this discrepancy: asked to the same sample, the ladder is better predicted by GDP than the satisfaction question, reproducing about half of the Gallup–WVS difference in predictive capacity, but both questions yield an income gradient close to Gallup's, whereas the WVS gradient is much flatter.

These findings have implications for research and policy. For research, cross-country conclusions about the income–well-being relationship should be checked across datasets and questions: the strength of this relationship depends on how and whom one asks, and the available evidence does not establish that one dataset is more representative than the other. The gap between datasets is largest in low-income countries, where neither the present experiment nor existing evidence can yet tell which survey is closer to the truth; replicating the experiment face-to-face in low- and middle-income countries is the natural next step. Other research avenues include testing whether region also predicts emotions better than income (Blanchflower & Bryson, 2024; Killingsworth et al., 2023), and, despite

evidence against translation issues (Diener & Suh, 2000), using anchoring vignettes to rule out response styles as an explanation of regional differences. For policy, the results suggest that absolute income is less determining for national well-being than is often thought: rich countries rarely have low well-being, but middle-income countries can be as happy as the richest, so that growth is neither necessary nor sufficient for high well-being. Well-being itself, rather than growth, its imprecise proxy, should be the primary target when evaluating reforms. And since non-material dimensions of life seem central to the regional differences in well-being, understanding their mechanisms deserves more attention (Frijters et al., 2020; Stiglitz et al., 2009).

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude Opus 5.5 to write analysis code and draft parts of the manuscript. After using this tool, the author reviewed and edited the code and content created by AI as needed and takes full responsibility for the content of the publication.

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A Additional results

Table A1: Correlation between national well-being indicators (WVS country-years).

	Happy	Very Happy	Very Unhappy	V. Happy – V. Unhappy	Happiness (mean)	Satisfaction (mean)	Satisfied	Happy + Satisfied
Happy	1.00	0.60	–0.73	0.70	0.89	0.73	0.72	0.90
Very Happy	0.60	1.00	–0.37	0.98	0.89	0.60	0.51	0.59
Very Unhappy	–0.73	–0.37	1.00	–0.54	–0.67	–0.58	–0.57	–0.69
V. Happy – V. Unhappy	0.70	0.98	–0.54	1.00	0.94	0.66	0.58	0.67
Happiness (mean)	0.89	0.89	–0.67	0.94	1.00	0.75	0.69	0.83
Satisfaction (mean)	0.73	0.60	–0.58	0.66	0.75	1.00	0.96	0.93
Satisfied	0.72	0.51	–0.57	0.58	0.69	0.96	1.00	0.95
Happy + Satisfied	0.90	0.59	–0.69	0.67	0.83	0.93	0.95	1.00

(Back to Section 2.1.)

Table A2: Slope of national well-being indicators with respect to $\log_{10}$ GDP per capita (PPP).

Data	Indicator	Slope	s.e.	<i>p</i> -value	R^2	Obs.
WVS	Happy	0.109	0.019	0.000	0.144	340
WVS	Very Happy	0.002	0.025	0.948	0.000	340
WVS	Very Unhappy	–0.022	0.004	0.000	0.072	340
WVS	V. Happy – V. Unhappy	0.023	0.026	0.379	0.004	340
WVS	Happiness (mean)	0.264	0.083	0.002	0.044	340
WVS	Satisfaction (mean)	1.143	0.142	0.000	0.233	340
WVS	Satisfied	0.236	0.024	0.000	0.316	340
WVS	Happy + Satisfied	0.173	0.020	0.000	0.275	338
Gallup	Satisfied	0.372	0.017	0.000	0.655	2344
Gallup	Very satisfied (8–10)	0.218	0.018	0.000	0.442	2344
Gallup	Extremely satisfied (9–10)	0.065	0.009	0.000	0.178	2344
Gallup	Completely satisfied (10)	0.012	0.005	0.008	0.015	2344
Gallup	Unsatisfied (0/1–4)	–0.304	0.012	0.000	0.639	2344
Gallup	Dissatisfied (0/1–2)	–0.135	0.009	0.000	0.432	2344
Gallup	Satisfaction (mean)	1.779	0.083	0.000	0.611	2344

Note: OLS across country-years; standard errors clustered by country. Gallup: 2006–2023. (Back to Section 3.2.)

Table A3: Out-of-sample predictive capacity: leave-one-country-out cross-validated R^2 .

Data	Indicator	Obs.	log GDP PPP	Best income variable	Region	log GDP PPP + Region
WVS, all waves	Happy	340	0.12	0.18	0.26	0.32
WVS, all waves	Very Happy	340	−0.03	−0.04	0.27	0.27
WVS, all waves	Very Unhappy	340	0.06	0.10	0.10	0.11
WVS, all waves	V. Happy – V. Unhappy	340	−0.02	−0.02	0.27	0.26
WVS, all waves	Happiness (mean)	340	0.02	0.05	0.29	0.30
WVS, all waves	Satisfaction (mean)	340	0.21	0.23	0.37	0.46
WVS, all waves	Satisfied	340	0.30	0.34	0.36	0.50
WVS, all waves	Happy + Satisfied	338	0.26	0.31	0.35	0.47
WVS, last obs. per country	Happy	115	0.05	0.03	0.11	0.12
WVS, last obs. per country	Very Happy	115	0.01	−0.05	0.17	0.16
WVS, last obs. per country	Very Unhappy	115	0.04	0.06	0.01	0.02
WVS, last obs. per country	V. Happy – V. Unhappy	115	−0.01	−0.08	0.16	0.15
WVS, last obs. per country	Happiness (mean)	115	−0.04	−0.09	0.12	0.10
WVS, last obs. per country	Satisfaction (mean)	115	0.18	0.19	0.30	0.33
WVS, last obs. per country	Satisfied	115	0.29	0.31	0.31	0.38
WVS, last obs. per country	Happy + Satisfied	115	0.24	0.25	0.28	0.34
Gallup, all years (2006–23)	Satisfaction (mean)	2344	0.61	0.66	0.49	0.68
Gallup, all years (2006–23)	Satisfied	2344	0.65	0.72	0.52	0.72
Gallup, all years (2006–23)	Unsatisfied (0/1–4)	2344	0.63	0.65	0.48	0.68
Gallup, last obs. per country	Satisfaction (mean)	167	0.60	0.58	0.45	0.64
Gallup, last obs. per country	Satisfied	167	0.65	0.68	0.52	0.70
Gallup, last obs. per country	Unsatisfied (0/1–4)	167	0.65	0.60	0.48	0.67
WHR, 2023–25 average	Satisfaction (mean)	147	0.62	0.60	0.52	0.69

Note: Each country's observations are predicted with a model estimated on all other countries. Cross-validated $R^2 = 1 - \sum(y - \hat{y}_{\text{out-of-sample}})^2 / \sum(y - \bar{y})^2$; it can be negative when a model predicts worse than the overall mean. In each row, the highest cross-validated R^2 among log GDP PPP, the best income variable and region is in bold. (Back to Section 3.3.)

Table A4: Robustness: income vs. region across specifications and datasets.

Specification	Obs.	Countries	R^2 income		R^2 region	Share of explained variance due to income			Region better
			log PPP	best		log PPP	best	adj. R^2	(share of cases)
WVS, all waves	340	112	0.14	0.20	0.33	0.23	0.33	0.27	0.98
WVS, waves 1–2 (1981–91)	26	21	0.07	0.43	0.58	0.09	0.40	0.10	1.00
WVS, wave 3 (1995–99)	55	55	0.16	0.33	0.65	0.14	0.28	0.18	1.00
WVS, wave 4 (1999–2004)	40	40	0.19	0.31	0.33	0.27	0.47	0.47	0.43
WVS, wave 5 (2004–09)	58	58	0.21	0.35	0.49	0.22	0.38	0.23	0.98
WVS, wave 6 (2010–16)	60	60	0.08	0.22	0.27	0.14	0.41	0.19	0.96
WVS, wave 7 (2017–23)	102	92	0.09	0.17	0.28	0.21	0.35	0.27	1.00
WVS, wave 7 w/o 2020–21	80	72	0.08	0.19	0.29	0.19	0.38	0.27	0.95
WVS, population-weighted	340	112	0.10	0.19	0.21	0.25	0.44	0.32	0.77
WVS, last obs. per country	115	112	0.13	0.19	0.25	0.26	0.39	0.30	0.88
WVS, no GDP imputation	321	108	0.13	0.20	0.32	0.22	0.33	0.27	0.98
WVS, GDP PPP 2017 \$ (old)	340	112	0.14	0.20	0.33	0.24	0.33	0.28	0.98
WVS, without EVS surveys	304	108	0.12	0.19	0.37	0.18	0.28	0.22	1.00
Gallup, all years (2006–23)	2344	167	0.42	0.48	0.42	0.46	0.51	0.49	0.31
Gallup, last obs. per country	167	167	0.42	0.44	0.41	0.44	0.48	0.45	0.37
Gallup, last obs., WVS countries	109	109	0.44	0.47	0.43	0.46	0.49	0.47	0.35
Gallup, last obs., 6 regions	167	167	0.42	0.44	0.42	0.43	0.46	0.44	0.41
Gallup, last obs., WB regions	167	167	0.42	0.44	0.38	0.47	0.50	0.48	0.29
WHR, 2023–25 average	147	147	0.63	0.63	0.55	0.56	0.56	0.56	0.00

Note: Averages over well-being indicators (8 for the WVS; 7 satisfaction indicators for Gallup; mean ladder for the WHR). “Best” refers to the income measure with the highest average R^2 in the main WVS specification (Cluster $k=7$ nominal). “Region better” is the share of indicator $\times$ income-measure combinations for which $s < 0.5$; values above 0.5 are in bold. Alternative region classifications and exclusions of regions: see Table 3. (Back to Section 3.3.)

Table A5: Other country-level correlates of national well-being: Shapley decomposition of R^2 (WVS).

Indicator	Other variable	Obs.	R^2 other alone	Shapley decomposition of R^2			
				Income	Region	Other	Total
Satisfaction (mean)	Freedom of choice	334	0.63	0.11	0.21	0.42	0.74
Satisfaction (mean)	Importance of God	329	0.01	0.17	0.33	0.02	0.51
Satisfaction (mean)	Tolerance (homosexuality)	326	0.24	0.12	0.30	0.10	0.52
Satisfaction (mean)	Importance of democracy	216	0.08	0.13	0.32	0.02	0.47
Satisfaction (mean)	GDP growth (5-year)	334	0.02	0.16	0.34	0.02	0.52
Happiness (mean)	Freedom of choice	333	0.39	0.02	0.22	0.27	0.51
Happiness (mean)	Importance of God	330	0.01	0.05	0.31	0.03	0.39
Happiness (mean)	Tolerance (homosexuality)	325	0.09	0.02	0.31	0.07	0.40
Happiness (mean)	Importance of democracy	216	0.03	0.02	0.28	−0.03	0.27
Happiness (mean)	GDP growth (5-year)	334	0.01	0.03	0.31	0.00	0.34

Note: Country-year averages of WVS variables: perceived freedom of choice and control (1–10), importance of God in one’s life (1–10), justifiability of homosexuality (1–10), importance of living in a democracy (1–10, waves 5–7), and average annual growth of real GDP per capita over the previous five years. Shapley decomposition of the R^2 of the regression on log GDP per capita, region dummies and the other variable. In each row, the largest contribution among income, region and the other variable is in bold. (Back to Section 3.5.)

Table A6: Within-country relation between national well-being and $\log_{10}$ GDP per capita (country fixed effects, WVS).

Indicator	Slope	s.e.	p -value	Within R^2	Obs.
Happy	0.286	0.043	0.000	0.281	312
Very Happy	0.207	0.048	0.000	0.146	312
Very Unhappy	−0.039	0.010	0.000	0.044	312
V. Happy – V. Unhappy	0.246	0.054	0.000	0.160	312
Happiness (mean)	1.063	0.180	0.000	0.245	312
Satisfaction (mean)	2.313	0.420	0.000	0.344	312
Satisfied	0.440	0.076	0.000	0.347	312
Happy + Satisfied	0.366	0.058	0.000	0.389	310

Note: Countries surveyed at least twice. Standard errors clustered by country. On average over the eight indicators, the within-country R^2 is 0.24. (Back to Section 3.5.)

Table A7: Countries most often ranked happiest (WVS, 8 indicators $\times$ 8 samples: each wave and all waves combined).

Country	Occurrences as happiest	Region
Switzerland	10	Western
Mexico	9	Latin America
Kyrgyzstan	6	Asia
Australia	5	Western
Netherlands	4	Western
Nigeria	4	Africa
Puerto Rico	4	Latin America
Colombia	3	Latin America
Finland	3	Western
Vietnam	3	Asia

Note: Looking at all waves combined, the happiest country-year is: Kyrgyzstan–2020 (3 indicators: Very Happy, V. Happy – V. Unhappy, Happiness (mean)); Finland–1981 (Satisfied); Malaysia–2012 (Very Unhappy); Mexico–2012 (Satisfaction (mean)); Qatar–2010 (Happy); Vietnam–2020 (Happy + Satisfied). For *Very Unhappy*, the happiest country-year is the one with the lowest share. (Back to Section 3.5.)

Table A8: Effects of question wording and scale on life evaluation, Fabre (2025) survey experiment.

Dependent Variables: Model:	Answer		Answer (0–10 equiv.)	Satisfied (6+)	Answer (0–10 equiv.)	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Ladder wording (Gallup)	-0.451*** (0.046)	-0.455*** (0.067)	-0.456*** (0.067)	-0.088*** (0.013)	-0.464*** (0.123)	-0.494** (0.172)
Scale 1–10	0.137** (0.046)	0.132 (0.068)	-0.260*** (0.072)	-0.008 (0.013)	-0.276*** (0.046)	-0.540** (0.182)
Ladder wording (Gallup) $\times$ Scale 1–10		0.008 (0.092)	-0.041 (0.097)	0.028 (0.019)		0.055 (0.247)
Income quartile (1–4)					0.689*** (0.031)	0.635*** (0.043)
Income quartile (1–4) $\times$ Ladder wording (Gallup)					-0.004 (0.042)	0.009 (0.058)
Income quartile (1–4) $\times$ Scale 1–10						0.106 (0.061)
Income quartile (1–4) $\times$ Ladder wording (Gallup) $\times$ Scale 1–10						-0.022 (0.083)
<i>Fixed-effects</i>						
Country	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	11,000	11,000	11,000	11,000	11,000	11,000
R ²	0.05681	0.05681	0.05942	0.04208	0.15976	0.16024

Heteroskedasticity-robust standard-errors in parentheses

Signif. Codes: ***, 0.001, **, 0.01, *, 0.05

Note: OLS with country fixed effects, survey weights (normalized within country) and heteroskedasticity-robust standard errors in parentheses. *Ladder wording:* Cantril ladder (vs. WVS life satisfaction). *Scale 1–10:* answers from 1 to 10 (vs. 0 to 10). In columns 3, 5 and 6, 1–10 answers are linearly stretched to 0–10. Columns 5 and 6 interact the treatments with the respondent’s income quartile within their country; in column 6, the interaction between income and wording is 0.009 ($p = .883$) and the triple interaction is -0.023 ($p = .786$). * $p < .05$, ** $p < .01$, *** $p < .001$. (Back to Section 4.3.)

Table A9: Decomposition of the WVS–Gallup gap by country.

Country	Year WVS/EVS (Gallup)	Survey: mode	Gallup (1)	WVS/EVS (2)	Gap D (2)–(1)	Ladder 0–10 (3)	Satisf. 1–10 (4)	Question effect Q (4)–(3)	Residual R $D - Q$
Poland	2017	EVS: CAPI	6.28	7.56	1.28	5.53	6.37	0.84	0.44
Spain	2017	EVS: CAPI	6.47	7.49	1.02	5.61	6.99	1.38	–0.36
Japan	2019	WVS: Mail	5.94	6.76	0.82	5.33	5.00	–0.33	1.15
Italy	2018	EVS: CAPI	6.69	7.28	0.59	5.99	6.59	0.60	–0.01
Germany	2018	WVS: CAPI	7.15	7.74	0.58	6.01	6.84	0.83	–0.25
United Kingdom	2022	WVS: CAPI/Web/Mail	6.76	7.32	0.56	5.67	6.77	1.10	–0.55
France	2018	EVS: CAPI	6.79	7.26	0.47	5.96	6.68	0.72	–0.25
Switzerland	2017	EVS: Web/Mail/CAPI	7.58	8.02	0.43	6.85	7.31	0.46	–0.03
Saudi Arabia	2003 (2006)	WVS: Face-to-face	7.06	7.28	0.23	6.31	7.10	0.79	–0.57
United States	2017	WVS: Web/Phone	7.23	7.22	–0.01	6.03	6.74	0.70	–0.71

Note: Mean answers on native scales (Gallup and ladder: 0–10; WVS/EVS and satisfaction: 1–10); columns (3) and (4): Fabre (2025). The Gallup year is given in parentheses when it differs from the WVS year. Survey: mode: source of the WVS/EVS data and main interview mode(s) (CAPI: computer-assisted personal interviews). (Back to Section 4.3.)

Table A10: Decomposition of the WVS–Gallup gap: summary.

Statistic	Estimate	95% bootstrap CI
Mean gap D (WVS – Gallup)	0.60	[0.55; 0.64]
Mean question effect Q (satisfaction 1–10 – ladder 0–10)	0.71	[0.56; 0.84]
Mean residual $R = D - Q$ (samples, mode, context)	–0.11	[–0.26; 0.04]
Share of mean gap due to question ($\bar{Q}/\bar{D}$)	1.19	[0.94; 1.45]
Share of gross level difference due to question ($(\bar{Q} /(\bar{Q} + \bar{R}))$)	0.86	[0.76; 0.99]
Share of cross-country variance of D due to Q ($\text{cov}(D, Q)/\text{var}(D)$)	0.11	[–0.33; 0.55]
Correlation between D and Q	0.09	[–0.26; 0.41]
Mean absolute gap $ D $	0.60	[0.56; 0.65]
Mean absolute residual $ R $	0.43	[0.33; 0.60]
Mean gap, share satisfied (6+)	0.07	[0.06; 0.08]
Mean question effect, share satisfied (6+)	0.11	[0.08; 0.14]
Variance share due to Q , share satisfied	0.04	[–0.33; 0.45]

Note: 10 countries. 95% confidence intervals from 1,000 bootstrap replications resampling respondents within country $\times$ variant (Fabre) and within country (WVS). For Gallup, only the distribution of answers is available: in each country, the numbers of respondents giving each answer (0 to 10) are redrawn from a multinomial distribution with the observed shares and sample size, which is equivalent to resampling respondents. (Back to Section 4.3.)

Table A11: Representativeness of WVS/EVS samples: sample shares relative to population benchmarks.

Measure	Surveys	Median				Correlation with log GDP
		All	GDP < 10k	10–30k	> 30k	
Tertiary education, 25+ (ratio sample/benchmark)	295	1.18	1.98	1.18	1.05	−0.40
Urban (ratio sample/benchmark)	76	1.44	2.19	1.45	1.22	−0.69
Aged 65+ (ratio sample/benchmark)	335	0.90	0.64	0.88	1.01	0.33
Women among married respondents − 0.5	337	0.01	−0.00	0.01	0.01	0.19

Note: One observation per WVS or EVS survey. Ratios: weighted sample share divided by the population share, medians across surveys (a ratio above 1 indicates over-representation). Benchmarks (World Bank WDI): share of the population aged 25+ with at least short-cycle tertiary education (UNESCO), or with complete or incomplete tertiary education (Barro & Lee, 2013) before 2011; urban population share (national definitions; sample: urban and peri-urban respondents, waves 4, 5 and 7 only); share aged 65+ among the population aged 15+ (sample: 65+ among respondents, usually 18+). Women among married: unweighted share of women among married or cohabiting respondents minus 0.5. Correlations with log GDP per capita (PPP) of the log ratios (or of the deviation). (Back to Section 4.4.)

Figure A1: Share of *Very happy* people vs. GDP per capita (PPP), WVS/EVS 1981–2023.

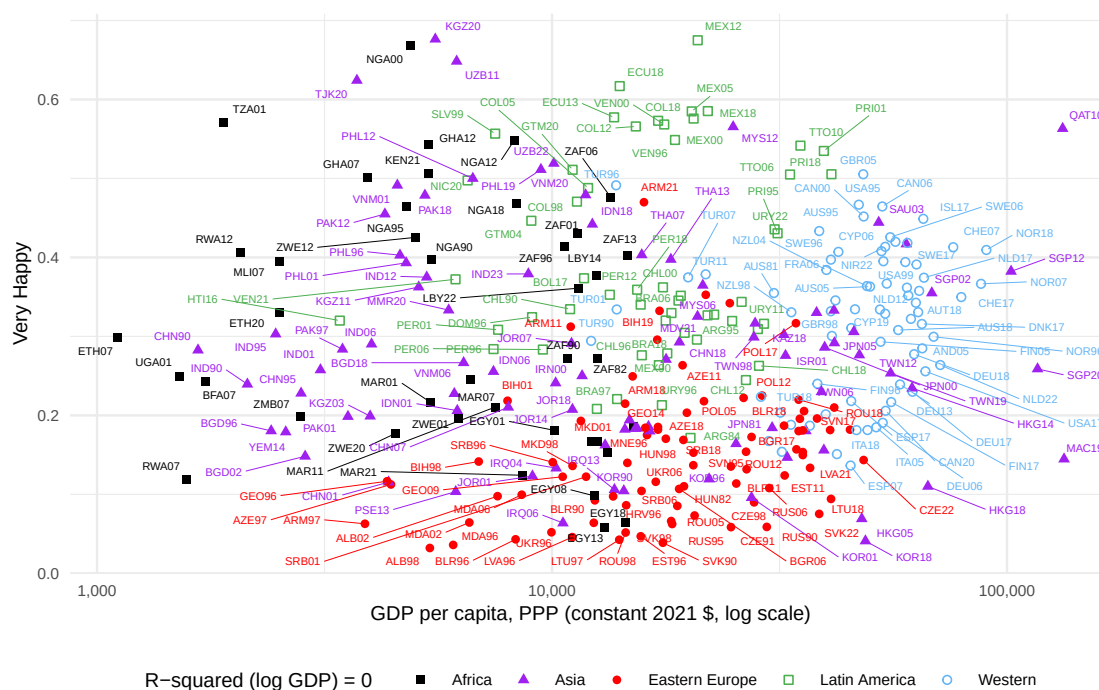


Figure A2: *Happy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.

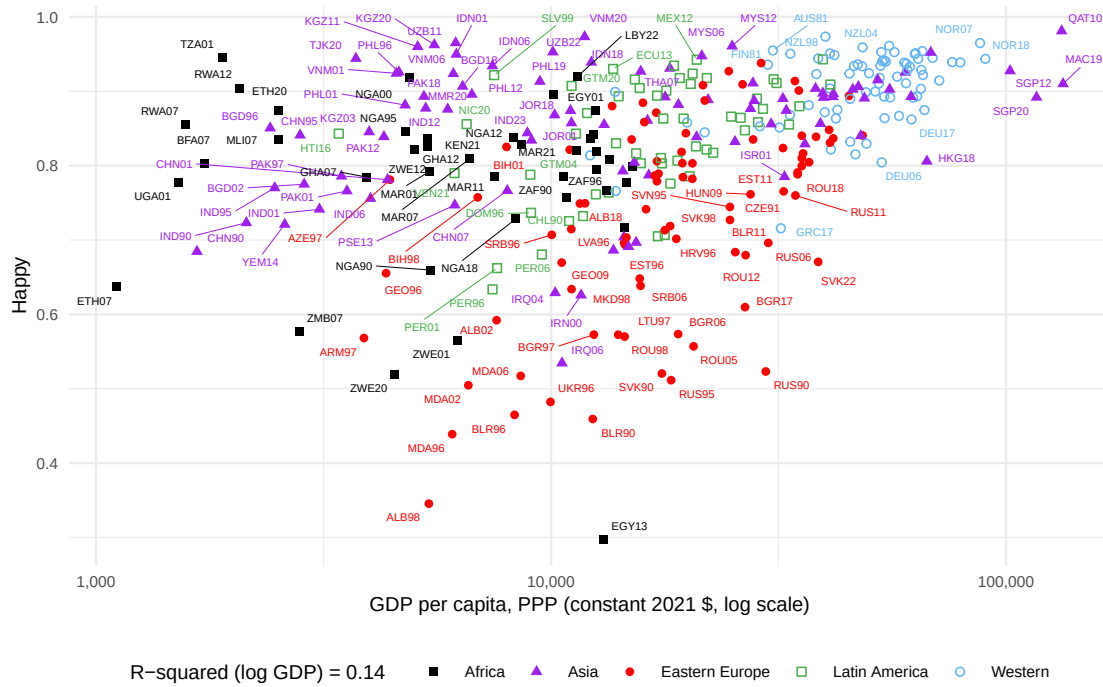
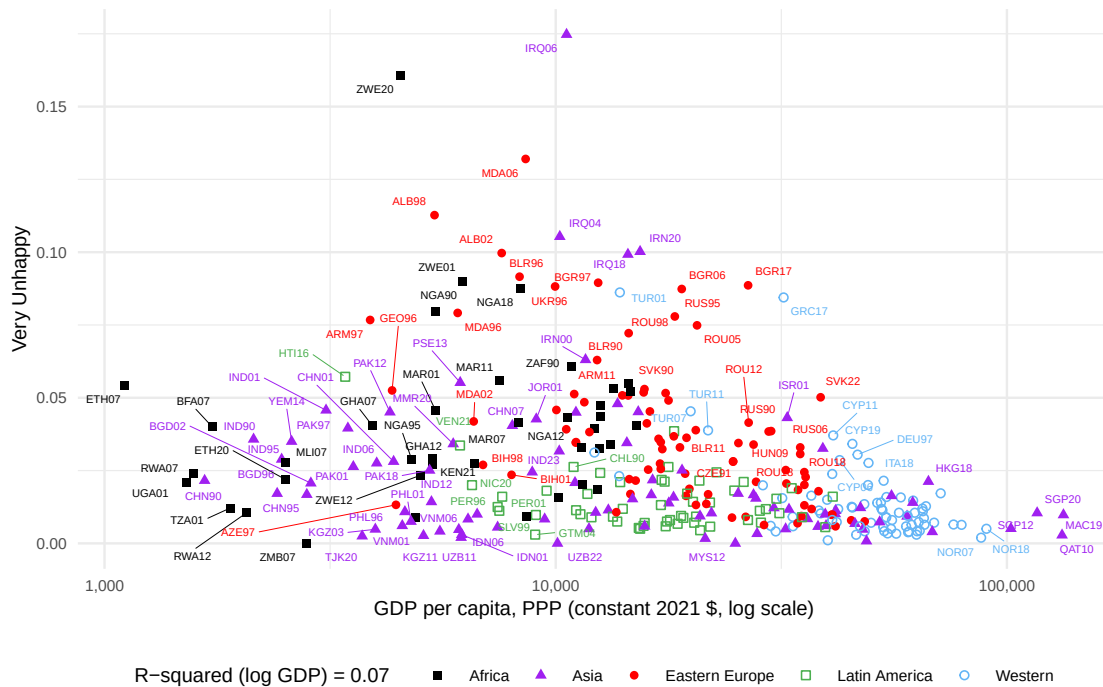


Figure A3: *Very Unhappy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.



Note: Egypt 2013 (42% of very unhappy people) is not shown, to keep the y -axis readable; the R^2 in the legend includes it.

Figure A4: *V. Happy* – *V. Unhappy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.

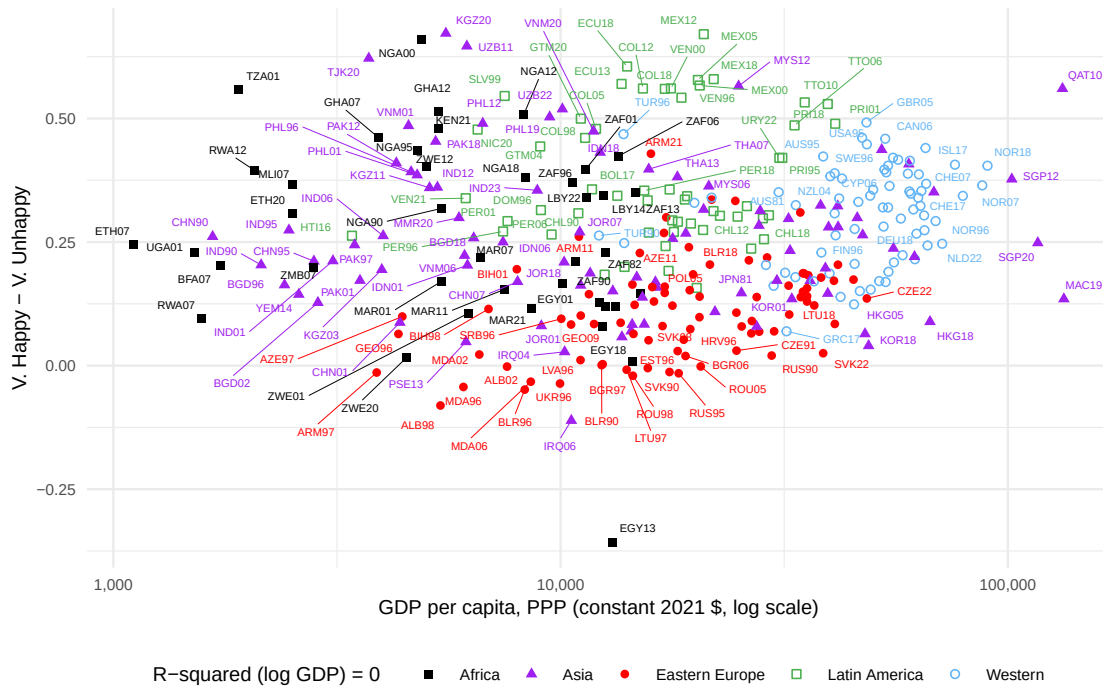


Figure A5: *Satisfied* vs. GDP per capita (PPP), WVS/EVS 1981–2023.

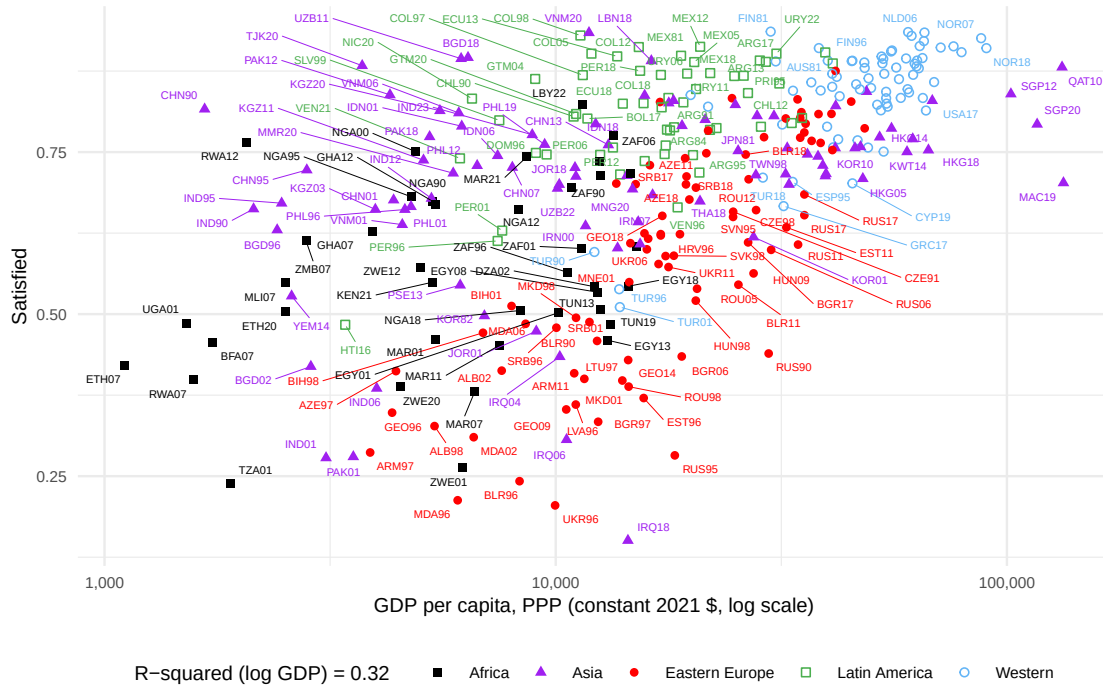


Figure A6: *Happy + Satisfied* vs. GDP per capita (PPP), WVS/EVS 1981–2023.

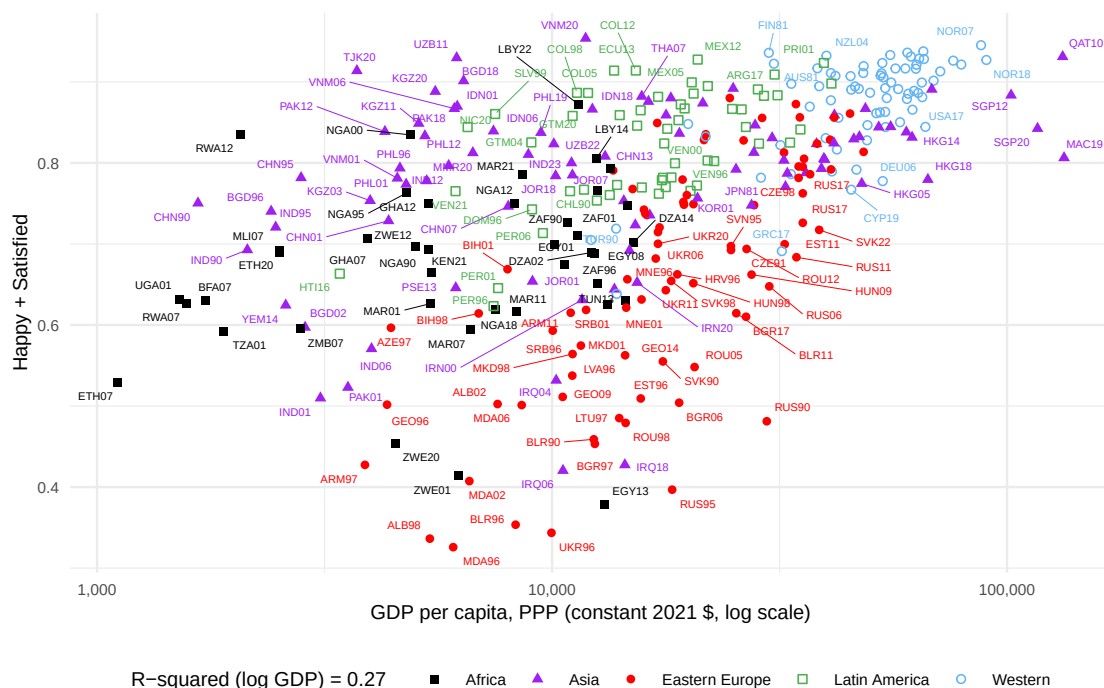
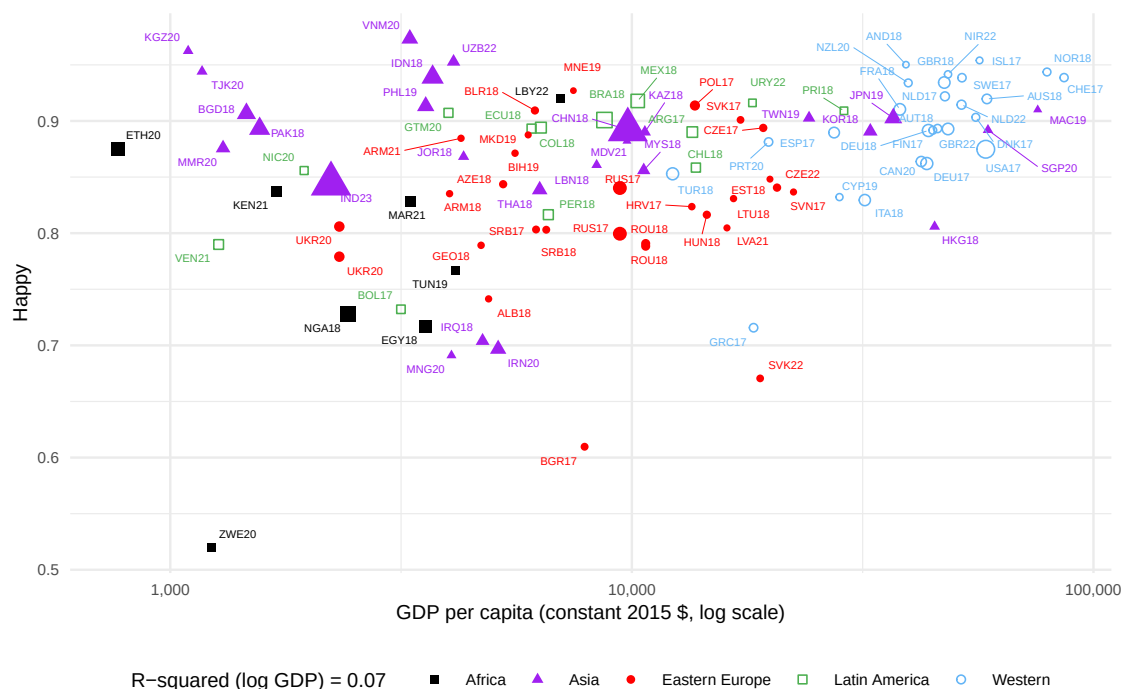


Figure A7: *Happy* vs. GDP per capita (nominal), WVS/EVS wave 7 (2017–2023), weighted by population.



Note: Point size and regression weights proportional to population.

(Back to Section 3.2.)